

Solution:

(a) Monte Carlo algorithm:

Randomly select a person A and accept A as a knight.

The probability of error is 0.4.

(b) Las Vegas algorithm:

1. Select a random person A .
2. Select a group of $0.8n$ inhabitants and ask each one of them whether A is a knight.
3. A is a knight if more than half of the group agree since there are at most $0.4n$ knaves.
4. Otherwise, repeat step 1 ~ 3 until we find a knight.

The possibility of finding a knight with exactly i rounds is $(0.4)^{i-1} \cdot 0.6$.

The expected number of questions is,

$$\begin{aligned}\mathbb{E}(\text{questions}) &= \sum_{i=1}^{\infty} (0.4)^{i-1} \cdot 0.6 \cdot i \cdot 0.8n \\ &= \sum_{i=0}^{\infty} (i+1) \cdot (0.4)^i \cdot 0.48n \\ &= \left(\frac{0.4}{(1-0.4)^2} + \frac{1}{1-0.4} \right) \cdot 0.48n \\ &\approx 1.333n < 1.5n\end{aligned}$$